A PROJECT REPORT ON
"RED CROSS SOCIETY — BLOOD DONATION MANAGEMENT SYSTEM"

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE 5TH SEMESTER OF
BACHELOR OF COMPUTER APPLICATIONS (B.C.A.)

SUBJECT: WEB FRAMEWORKS & SERVICES (WFS)

SUBMITTED BY:

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UNDER THE GUIDANCE OF:

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SUBMITTED TO:

Department of Computer Applications

College of Applied Science and Professional Studies, Chikhli

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ACADEMIC YEAR: 2026 – 2027

COLLEGE OF APPLIED SCIENCE AND PROFESSIONAL STUDIES, CHIKHLI

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CERTIFICATE

This is to certify that **Mr./Ms. [Your Full Name]**, Exam Seat No: **[Your Seat No]**, student of **B.C.A. 5th Semester**, has successfully completed the project work entitled **"RED CROSS SOCIETY — BLOOD DONATION MANAGEMENT SYSTEM"** in the subject of **Web Frameworks & Services (WFS)** as part of the curriculum prescribed for the award of the degree of Bachelor of Computer Applications (B.C.A.) for the academic year 2026–2027.

Date: 13/10/2026
Place: Chikhli / Navsari

Project Guide	Head of Department	External Examiner

DECLARATION

I, the undersigned, hereby declare that this project report entitled **"RED CROSS SOCIETY — BLOOD DONATION MANAGEMENT SYSTEM"** submitted in partial fulfillment of the requirements for the 5th Semester of Bachelor of Computer Applications (B.C.A.) is an authentic record of independent project work carried out by me under the supervision of my project guide.

The conceptual designs, application source code architecture, database schemas, and documentation presented in this report are original. Any external standard or technical reference utilized (such as the e-RaktKosh National Transfusion Guidelines) has been duly acknowledged.

I further declare that this report has not been submitted previously to any other university or institution for the award of any degree or diploma.

Date: 13/10/2026
Place: Navsari

[Your Full Name]
(T.Y. B.C.A. - 5th Semester)

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Date: 13/10/2026
Place: Navsari

[Your Full Name]
(T.Y. B.C.A. - 5th Semester)

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1. Introduction

The **Red Cross Society — Blood Donation Management System** is an enterprise-oriented web application designed to manage, streamline, and centralize the operational workflow of regional blood banks, voluntary donor drives, and emergency transfusion requirements. Built on modern web architecture using PHP (Laravel Framework) and MySQL, the application draws its design principles directly from the national blood transfusion portal, **e-RaktKosh**.

In life-saving situations, locating verified, matching blood components (Whole Blood, PRBC, Platelet Concentrate, Fresh Frozen Plasma, and Cryoprecipitate) is time-critical. Traditional, localized blood registries rely on manual paper logging, unscheduled physical visits, and uncoordinated emergency calls. This lack of centralized visibility often leads to critical delays in patient care.

This system bridges that gap by offering a transparent, searchable, and secure public platform. The system supports three primary user roles:

- **Voluntary Donors:** Can check their clinical donation intervals, locate upcoming blood donation camps, and register for digital donor certificates.
- **Hospital Attendants & Emergency Recipients:** Can check real-time regional stock and submit urgent patient requisitions directly to the central blood bank.
- **Blood Bank Transfusion Officers (Admin):** Can manage barcoded unit inventories, verify clinical physical screening vitals, and approve blood requests through a multi-stage clinical workflow.

By applying Model-View-Controller (MVC) architectural separation and Role-Based Access Control (RBAC), the project delivers high performance, secure session management, and relational data integrity.

1.1 Existing System

In traditional and paper-bound blood bank centers, operations rely heavily on manual recordkeeping and decentralized communication:

- **Manual Registry Logbooks:** Blood units collected from donors are recorded by hand in physical paper registers. Staff must manually record donor contact information, collection dates, and component division logs.
- **Physical Counter Enquiries:** Family members and hospital attendants must visit individual blood banks or call multiple facilities to verify whether specific blood groups are available in stock.
- **Paper-Based Pre-Donation Forms:** Donors fill out physical medical intake forms to record preliminary physical checks (weight, hemoglobin, blood pressure, pulse). These paper records are prone to wear, physical misplacement, or accidental destruction over time.
- **Uncoordinated Emergency Requests:** When an emergency occurs at a hospital, physical requisition slips signed by doctors are carried between hospital departments and blood centers, resulting in administrative bottlenecks during acute emergencies.

1.2 Limitations of the Existing System

The manual blood management process presents several operational challenges for both medical staff and patients:

1. **Time-Consuming & High Operational Latency:** Manual ledger checks increase wait times for life-saving component transfusions.
2. **Absence of Centralized Inventory Visibility:** Attendants cannot review regional multi-center inventory from a single interface.
3. **Risk of Data Loss & Wear:** Paper records deteriorate, take up significant storage space, and cannot be backed up securely in the cloud.
4. **Vulnerability to Clinical Oversights:** Manual methods can miss minimum donation cooldown periods (90 days for males and 120 days for females) or fail to flag expiring blood bags in time.
5. **Lack of Role-Based Security:** Paper logs lack password encryption, access control checks, or tamper-proof audit trails for inventory adjustments.

2. Proposed System

The proposed web application replaces manual logging with a secure, centralized portal. Designed around the public health standards of **e-RaktKosh**, it automates live inventory updates, donation intervals, and medical requisitions.

- **Live Inventory Tracking:** Displays cleared, unexpired blood bags across all major blood types (A+, A-, B+, B-, AB+, AB-, O+, O-) as well as rare groups like the Bombay Phenotype.
- **Barcoded Unit/Batch Identification:** Tracks individual bags using unique serial barcodes (`unit_barcode`), recording bag capacity (350ml or 450ml), mandatory screening clearance (HIV, Hepatitis B/C, Malaria, Syphilis), and shelf-life expiration dates.
- **Algorithmic Cooldown Engine:** Enforces gender-based clinical recovery intervals (90 days for male donors, 120 days for female donors) before allowing donation scheduling.
- **Digital Donor Identification:** Generates a unique, trackable donor certificate reference (`CERT-RCS-XXXXXX`) for verified donors.
- **Multi-Stage Hospital Requisition Pipeline:** Moves patient requests systematically across clinical stages: Pending → Cross-Match Verified → Approved / Issued → Fulfilled / Rejected.
- **Role-Based Access Control (RBAC):** Restricts administrative functions (such as stock issuance and workflow changes) to authorized staff via encrypted login credentials.

2.1 Project Profile

Attribute	Project Specification
Project Title	Red Cross Society — Blood Donation Management System
Academic Program	Bachelor of Computer Applications (B.C.A.) – Semester 5
Course / Subject	Web Frameworks & Services (WFS)
System Architecture	Model-View-Controller (MVC)
Backend Framework	Laravel 12 (Core PHP 8.2+)
Frontend Technologies	Blade Engine, HTML5, CSS3, JavaScript, Bootstrap 5.3
Database Server	MySQL via phpMyAdmin (XAMPP Environment)
Web Server Environment	Apache HTTP Server 2.4 (Configured Port 8080)
Security Standards	Bcrypt Password Hashing, Session Regeneration, RBAC Middleware
Target User Groups	Public Donors, Hospital Transfusion Units, Blood Bank Admins

2.2 Objectives

1. **Digitize Blood Bank Records:** Replace paper logs with structured MySQL relational tables for inventory, donor history, and requisitions.
2. **Provide Real-Time Public Stock Visibility:** Offer an accessible search interface where hospital staff and attendants can filter blood units by group, component, and district.
3. **Enforce Clinical Safety Standards:** Implement automated server-side checks for age (18–65), vital metrics, and mandatory post-donation cooldown intervals.
4. **Streamline Emergency Requests:** Enable hospital attendants to submit digital blood requisitions, reducing the need for physical travel during emergency cross-matching.
5. **Secure Administrative Functions:** Use dedicated middleware to restrict inventory approvals, state transitions, and drive scheduling to authorized medical officers.

2.3 Hardware and Software Requirements

Hardware Requirements

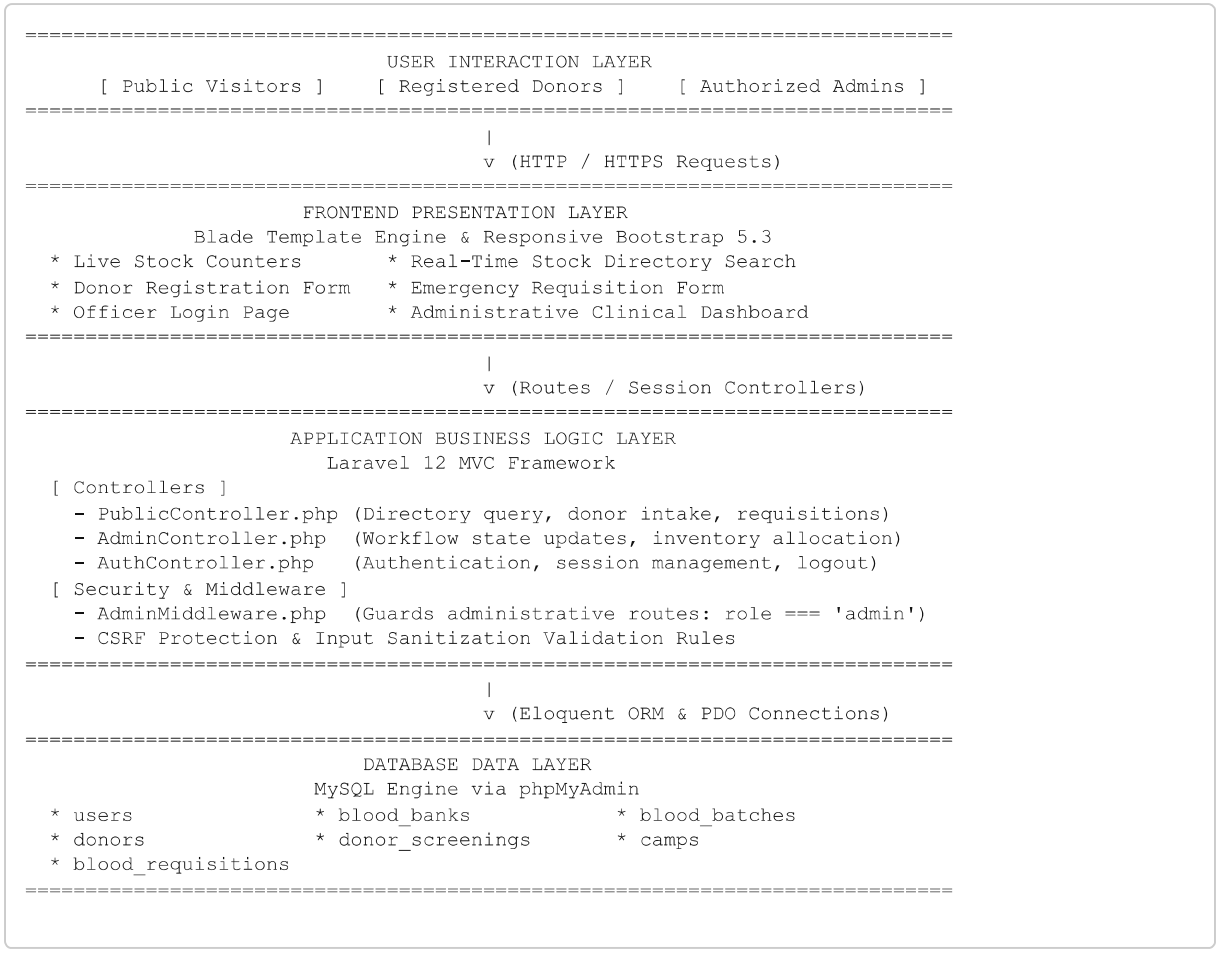
- **CPU:** Dual-core 2.0 GHz or higher (Intel Core i3/i5 or AMD Ryzen 3/5 equivalent)
- **RAM:** Minimum 4 GB (8 GB recommended for simultaneous server hosting)
- **Hard Disk Storage:** Minimum 500 MB free local disk space
- **Display:** 13-inch or higher screen with a minimum resolution of 1366 x 768

Software Requirements

- **Operating System:** Windows 10 / Windows 11 (64-bit)
- **Local Web Server:** Apache 2.4 (via XAMPP suite)
- **Database Management System:** phpMyAdmin / MySQL 8.0+
- **Programming Environment:** PHP 8.2.12 or higher
- **Framework Engine:** Laravel 12.x
- **IDE / Development Editor:** Visual Studio Code
- **Web Browser:** Google Chrome / Mozilla Firefox

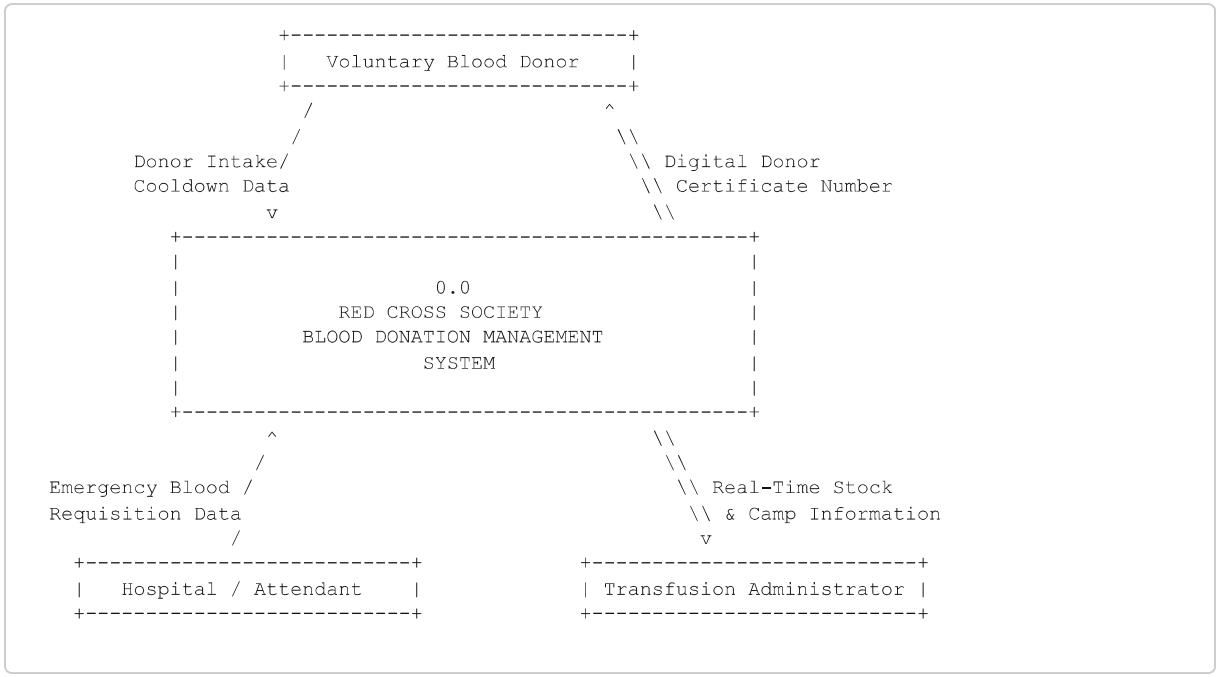
3. System Design

3.1 System Level Diagram (SLD)



3.2 Data Flow Diagrams (DFD)

Context Level Diagram (DFD Level 0)



DFD Level 1 (Detailed Subsystem Data Flow)

```
[ DONOR ]
    |---> Enters Profile, Gender, and Last Donation Date
    v
( 1.0 Donor Cooldown Evaluation Engine )
    |---> Calculates gender interval (Male: 90 days / Female: 120 days)
    |---> Writes verified record to [ donors ] table
    |---> Issues unique certificate identifier: CERT-RCS-XXXXXX

[ HOSPITAL RECIPIENT ]
    |---> Submits Patient Vitals, Diagnosis, Group, and Urgency
    v
( 2.0 Requisition Intake Subsystem )
    |---> Assigns initial workflow status: 'Pending'
    |---> Writes new record to [ blood_requisitions ] table

[ GENERAL PUBLIC ]
    |---> Selects Blood Group, Component, and District filters
    v
( 3.0 Live Inventory Search Engine )
    |---> Reads verified, unexpired units from [ blood_batches ] table
    |---> Renders dynamic availability table with unique barcodes

[ TRANSFUSION ADMIN ]
    |---> Enters email and password credentials
    v
( 4.0 Authentication & Session Middleware )
    |---> Checks credentials and role against [ users ] table
    |---> Grants access to the administrative control panel
    v
( 5.0 Clinical Workflow Management )
    |---> Updates status: 'Pending' -> 'Approved / Issued'
    |---> Automatically deducts matching units in [ blood_batches ] table
    |---> Records changes in [ blood_requisitions ] table
```

3.3 Data Dictionary

Table 1: users (Authentication & Role Credentials)

Column Name	Data Type	Null	Key	Description
id	BIGINT UNSIGNED	No	PK	Auto-increment primary key
name	VARCHAR(255)	No	-	Full name of registered user or staff officer
email	VARCHAR(255)	No	UNI	Login email address
phone	VARCHAR(20)	Yes	-	10-digit mobile number
password	VARCHAR(255)	No	-	Bcrypt-hashed password string
role	ENUM	No	-	'admin', 'donor', 'hospital' (Default: 'donor')
remember_token	VARCHAR(100)	Yes	-	Persistent login token string
created_at / updated_at	TIMESTAMP	Yes	-	Standard timestamps

Table 2: blood_banks (Regional Facility Directory)

Column Name	Data Type	Null	Key	Description
id	BIGINT UNSIGNED	No	PK	Primary key
name	VARCHAR(255)	No	-	Official blood bank name
license_number	VARCHAR(100)	No	UNI	Drug control license number
category	ENUM	No	-	'Govt', 'Red Cross', 'Charitable', 'Private'
district / city_taluka	VARCHAR(100)	No	-	Territorial jurisdiction
pincode	VARCHAR(10)	No	-	6-digit postal code
contact_person / helpline_phone	VARCHAR	No	-	Nodal contact credentials
has_component_facility	TINYINT(1)	No	-	Component separation capacity flag

Table 3: blood_batches (Individual Unit Barcode Inventory)

Column Name	Data Type	Null	Key	Description
id	BIGINT UNSIGNED	No	PK	Primary key
blood_bank_id	BIGINT UNSIGNED	No	FK	References blood_banks(id)
unit_barcode	VARCHAR(100)	No	UNI	Unique unit barcode (e.g., 'RCS-1001')
blood_group	ENUM	No	-	A+, A-, B+, B-, AB+, AB-, O+, O-, Bombay Phenotype
component	ENUM	No	-	'Whole Blood', 'PRBC', 'FFP', 'Platelet Concentrate'
bag_volume	ENUM	No	-	'350ml', '450ml'
collection_date / expiry_date	DATE	No	-	Shelf-life lifecycle dates
testing_clearance_status	ENUM	No	-	'Pending', 'Cleared', 'Failed'
status	ENUM	No	-	'Available', 'Reserved', 'Issued', 'Discarded'

Table 4: donors (Registered Donors & Intervals)

Column Name	Data Type	Null	Key	Description
id	BIGINT UNSIGNED	No	PK	Primary key
user_id	BIGINT UNSIGNED	Yes	FK	References users(id)
full_name	VARCHAR(100)	No	-	Full legal name
gender	ENUM	No	-	'Male', 'Female', 'Other'
dob	DATE	No	-	Date of birth (18-65 years)
phone	VARCHAR(15)	No	-	10-digit mobile number
blood_group	ENUM	No	-	Donor blood group
last_donation_date	DATE	Yes	-	Most recent donation date
is_eligible	TINYINT(1)	No	-	Cooldown eligibility status
certificate_number	VARCHAR(100)	Yes	UNI	Unique certificate reference

Table 5: donor_screenings (Clinical Mini-Physical Evaluations)

Column Name	Data Type	Null	Key	Description
id	BIGINT UNSIGNED	No	PK	Primary key
donor_id	BIGINT UNSIGNED	No	FK	References donors(id)
weight_kg	DECIMAL(5,2)	No	-	Body weight (Requirement: >= 45.0 kg)
hemoglobin_g_dl	DECIMAL(4,1)	No	-	Hb level (Requirement: >= 12.5 g/dL)
pulse_rate	INT	No	-	Measured pulse rate (60 - 100 bpm)
blood_pressure	VARCHAR(20)	No	-	Measured BP reading (e.g. '120/80')
screening_result	ENUM	No	-	'Fit', 'Temporary Deferral', 'Permanent Deferral'

Table 6: camps (Blood Donation Drives)

Column Name	Data Type	Null	Key	Description
id	BIGINT UNSIGNED	No	PK	Primary key
camp_title	VARCHAR(255)	No	-	Official title of the drive
organizing_institution	VARCHAR(255)	No	-	Partner NGO / College institution
venue_address	TEXT	No	-	Physical venue location
camp_date	DATE	No	-	Scheduled date
start_time / end_time	TIME	No	-	Operating drive timing

Table 7: blood_requisitions (Hospital Emergency Requests)

Column Name	Data Type	Null	Key	Description
id	BIGINT UNSIGNED	No	PK	Primary key
patient_name / age	VARCHAR / INT	No	-	Patient credentials
admitted_hospital	VARCHAR(255)	No	-	Receiving hospital facility
blood_group / component	ENUM	No	-	Required transfusion unit
units_required	INT	No	-	Quantity of units requested
urgency_level	ENUM	No	-	'Elective', 'Emergency', 'SOS'
workflow_state	ENUM	No	-	'Pending', 'Cross-Match Verified', 'Approved / Issued'